

UCLA Study Spots Recommendation System

Kathy Mo

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App: <https://stat418-kathy-final-project-bmpv6hxtgtpelge29ujkqn.streamlit.app/>

API: <https://ucla-study-api-853468627492.us-central1.run.app/docs>

Project Overview and Motivation

- Problem: Students may have problems knowing where to study
- This API and Streamlit app gives the user study spot recommendations
- Lists 30 different study spots on campus
- Has a short description of each spot, along with a link to its website
- Great for finding study spots for more variety in settings
- Can recommend and encourage students to try different spots on campus

Data Collection and Preprocessing

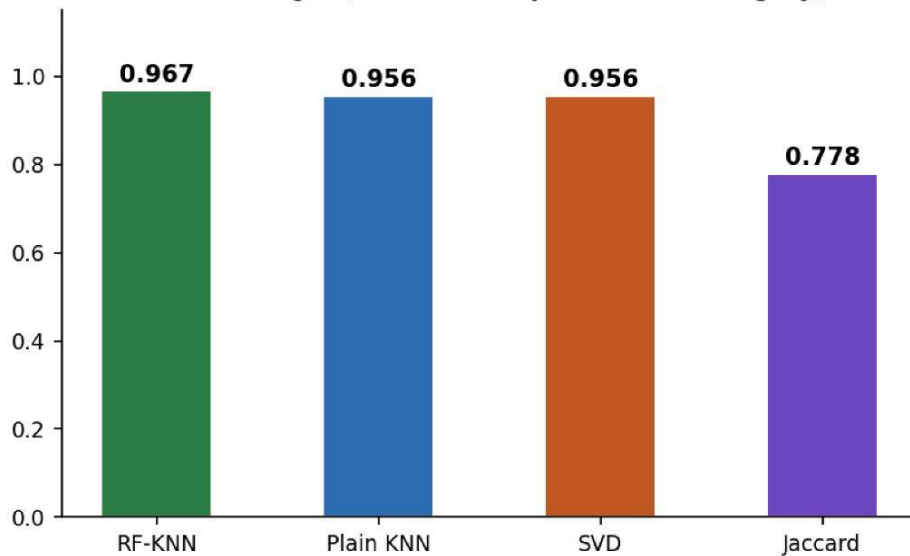
- Web scraped and got information from several different websites
 - Include the UCLA library and dining website
- Feature engineering:
 - Encoded noise level as silent all the way to moderate
 - Created scores for comfort, connectivity, and group-friendliness
- One-hot encoded feature tags for model
 - Such as whether spot has wifi or not
- Final data set has 22 columns, with features such as name, noise_level
- Dataset had high dimensionality, so removed the 15 feature tags that appeared once

Model Development and Evaluation

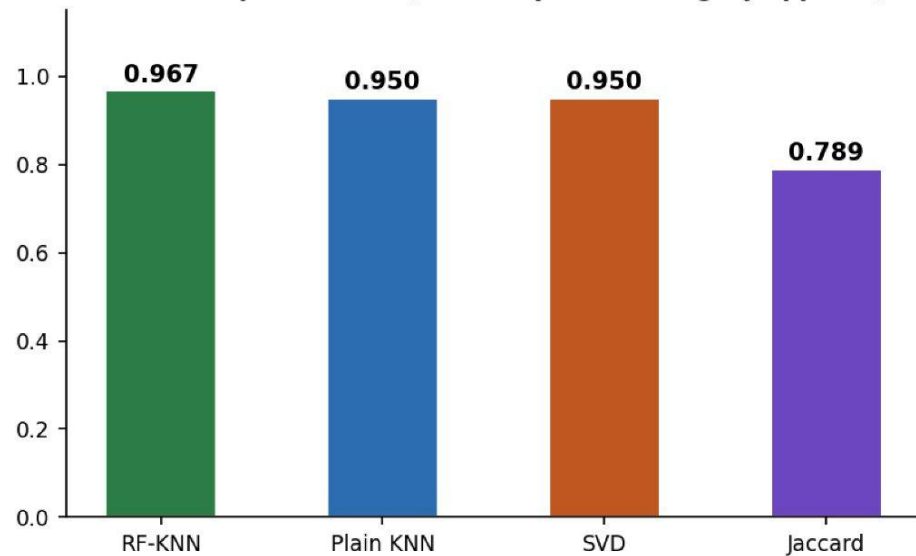
- Model aims to filter spots, not classify or predict
- Compared 4 different models:
 - RF-Weighted KNN: RF learns which features best distinguish spot types, then KNN uses those weights to find the most similar spots to user preferences
 - KNN: finds the nearest spots using cosine similarity, treating all features equally
 - SVD: finds similar spots based on shared underlying patterns in the feature matrix
 - Jaccard: measures overlap between binary feature tags (wifi, outlets, quiet, etc.) across spots, no weighting
- Evaluated using Precision@3 and Mean Reciprocal Rank (MRR)
 - Precision@3: what fraction of the top 3 results are relevant?
 - MRR: how early does the first relevant result appear?
- Best model is Random Forest weighted KNN
 - Precision@3 = 0.967 and MRR = 0.967
 - Top features: WiFi presence, outdoor/indoor, noise level

Model Comparison — Study Spot Retrieval Quality

Precision @ 3 (fraction of top-3 in same category)

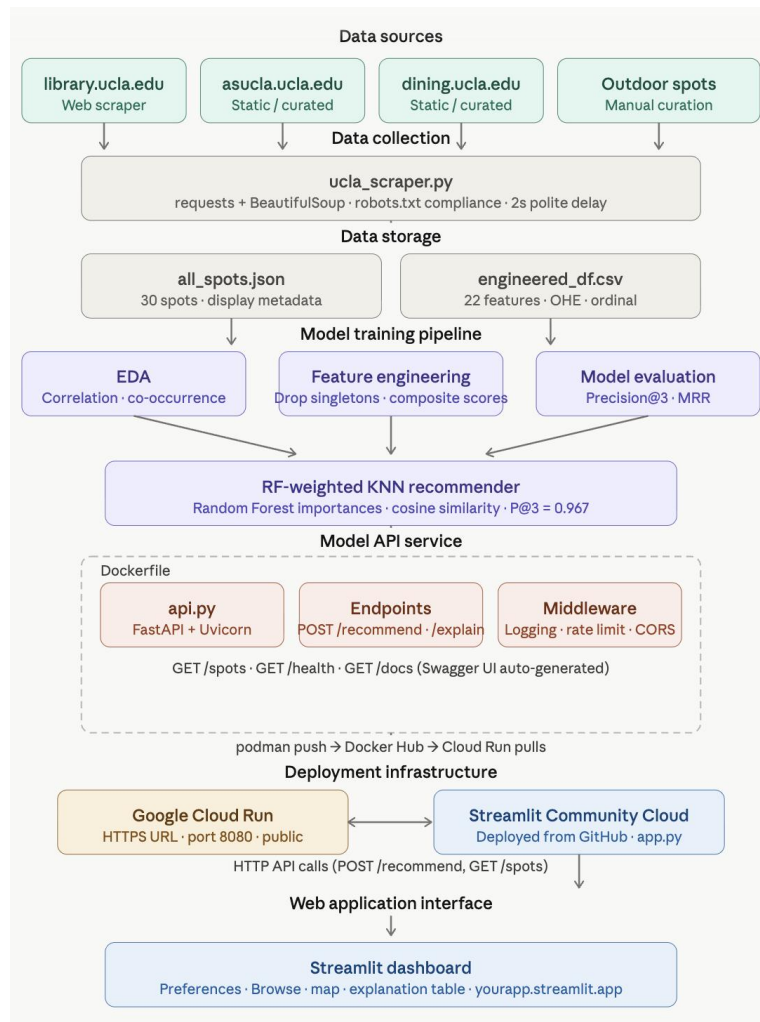


Mean Reciprocal Rank (how early same-category appears)



Architecture Diagram

- Two separate deployments: FastAPI on Google Cloud Run, Streamlit app on Streamlit Community Cloud
- API containerized with Docker, pushed to Docker Hub, pulled by Cloud Run
- Streamlit app deployed directly from GitHub



Application features, UI, and Walkthrough

☐

Coffee nearby

☐

Food nearby



Environment

☐

Need it quiet

☒

Prefer outdoor

☐

Comfortable seating



Session Type

☐

Group study



Other

☒

OK with weather-dependent spots

Noise tolerance

Quiet

Silent

Moderate

How critical is connectivity?

Nice to have

Not at all

Critical

Number of results

3

1

10



Find Study Spots

Recommendations

Browse All Spots

About

Your Top Study Spots

Showing 3 recommendations based on your preferences.

1.



Ackerman Union Outdoor Terrace

Outdoor · Central Campus

Outdoor seating on the Ackerman Union terrace overlooking Ackerman Square. Shaded tables. Close to printing and food options inside the building.

Outdoor

Shaded

Central Location

Printing Nearby

Noise level:  Moderate

Best times:

-  mornings before 10am
-  late afternoon after 4pm

Avoid:

-  noon-2pm on weekdays

Food nearby: Multiple options inside Ackerman Union

 Google Maps

Why was this recommended?

2.



Court of Sciences

Outdoor · South Campus

Open plaza surrounded by engineering and science buildings. Benches and tables. Adjacent to Boelter Hall, Young Hall, and the SEL.

Match

74%

Match

72%

Fork



Application features, UI, and Walkthrough

[Recommendations](#) [Browse All Spots](#) [About](#)

All UCLA Study Spots

Filter by category

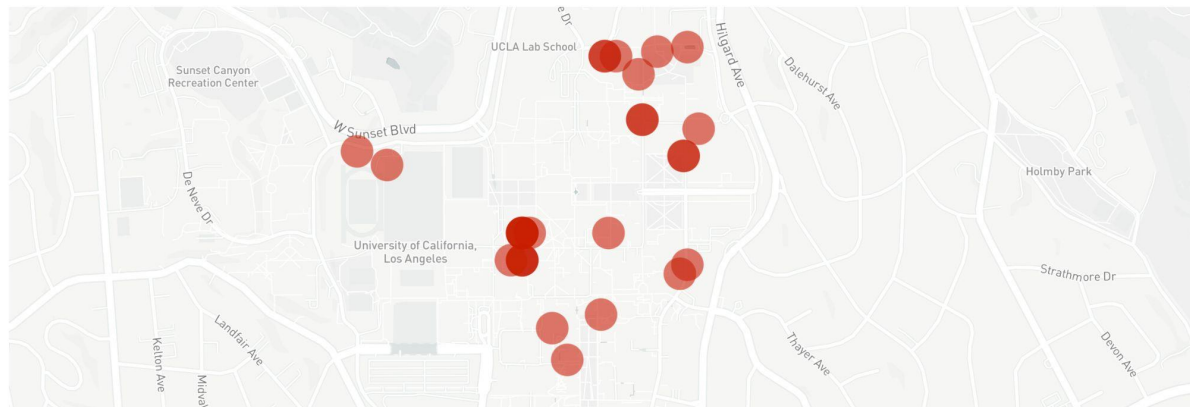
[Library](#) [Study Lounge](#) [Cafe](#) [Outdoor](#)

Filter by noise level

[Silent](#) [Very Quiet](#) [Quiet](#) [Low-Moderate](#) [Moderate](#)

- ☐ WiFi only
- ☐ Outlets only
- ☐ Outdoor / weather-independent only

Showing 30 of 30 spots



Challenges and Lessons Learned

- ASUCLA and dining websites are JavaScript-rendered, so web scraping returns empty content
 - Data had to be curated manually
- AI-generated code needs to be tested
 - Can hallucinate or give seemingly correct answers
- AI can be great at teaching how to deploy APIs, but it won't always be right
 - There was lots of debugging
- Used AI assistants to help create app user interface
- To improve, could have data on how busy each study spot is, if available
- Future work would add more temporal data